CANDIDATE NAME: ____

CENTRAL EAST EDUCATION DIVISION

2025 MSCE MOCK EXAMINATION

MATHEMATICS

Friday, 28 March

Subject Number: M131/II
Time Allowed: $2\frac{1}{2}$ hours
08:00 – 10:30 am

PAPER II

(100 marks)

Instructions

1. **This paper contains 16 pages. Please Check.**
2. Answer all the **six** questions in section **A** and any **four** in section **B**.
3. The maximum number of marks for each answer is indicated against each question.
4. Write your answers in the spaces provided on the question paper.
5. Scientific calculators may be used.
6. **All working must be clearly shown.**
7. Write your name on top of each page of this paper

Question Number	Tick if answered	Do not write in these columns	
1			
2			
3			
4			
5			
6			
7			
8			
9			
10			
11			
12			

Section A (60 marks)

Answer all the **six** questions in this section in the spaces provided.

1. a). Simplify $\frac{x-a}{x+a} - \frac{x+a}{x-a}$. **(4 marks)**

b) Without using a calculator simplify $\frac{\sqrt{2}+2\sqrt{5}}{\sqrt{5}-\sqrt{2}}$ giving the answer with a rational denominator. **(5 marks)**

2. a. Suppose that triangle ABC has vertices A (2, 5), B (3,3) and C (4,5). Show that triangle ABC is isosceles. **(5 marks)**

2 b. Prove that the angle which the chord makes with the tangent is equal to the angle subtended by the same chord in the alternate segment.
(6 marks)

3 a. Figure 1 is a circle ABCD and DE is a tangent to the circle at D. AC is parallel to tangent DE.

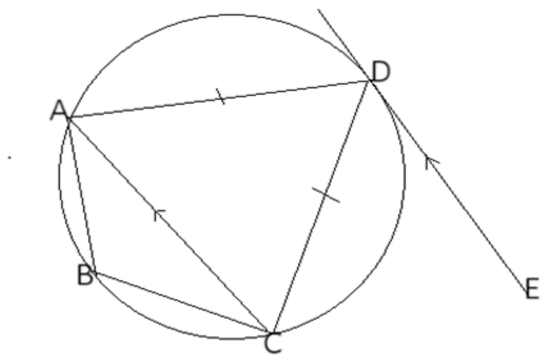

Figure 1

Prove that angle ABC is twice angle DAC. (6 marks)

3. b. A rectangular garden has a perimeter of 40 meters. If its area is $91m^2$, calculate the length of the garden. (**5 marks)**

4. a. Show that points L (4,0), M (14,11) and N (19, 16.5) are collinear.
(4 marks)

4. b. Solve the equation 5^{x+1} - 3 × 5^{x-1} = 22. (5 marks)

5. a. Figure 2 is a triangle KLM and KL = KM.

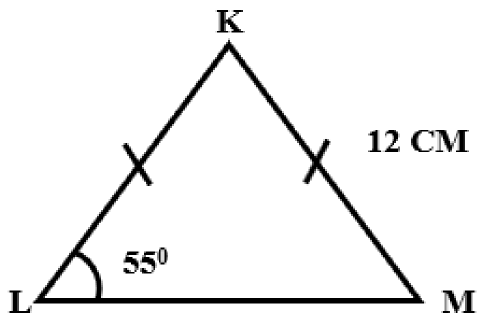

Figure 2

If KM = 12cm, calculate the length of LM giving the answer correct to the nearest whole number. (5 marks)

5 b. Using a ruler and a pair of compasses only construct in the same diagram:

- A circle center O with diameter 5cm.
- A point B outside the circle and 9cm from the center
- A tangent AB at A. (5 marks)

6 a. Kondwani has two fair spinners, spinner A has numbers 1,2,3. Spinner B has numbers 3,4,5.6. If Kondwani spins both spinners,

i. Draw probability space to show the sum of all the possible total scores. (3 marks)

ii. Find the probability that Kondwani's total score is 7. (2marks)

6 b. Given that the total surface area of a hemisphere is 1848 cm^2. Calculate the diameter of the hemisphere. (5 marks)

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SECTION B (40 marks)

Answer any **four** questions from this section in the spaces provided.

6. Table 1 shows values of x and y for the equation $y = x(x-1)(x-3)$.

Table 1

X	***-2***	***-1***	***0***	***1***	***2***	***3***	***4***
Y	***-30***		***0***	***0***		***0***	***12***

a. Complete the values of y in the table. (2 marks)

b. Using the scale of 2cm to represent 1 unit on horizontal axis and 2cm to represent 5 units on the vertical axis, draw the graph of $y = x(x-1)(x-3)$ on the graph paper provided. (4 marks)

c. Use the graph to solve the equations $y = x(x-1)(x-3)$ and $y = 2x - 6$ simultaneously.

(4 marks)

8 A man left city X and drove on a bearing of 047° to city Y. He then drove 75 km on a bearing of 117° to city Z which is on a bearing of 100° from city X. calculate the distance between cities X and Y, giving your answer correct to two decimal places. (10 marks)

9 Solve the equation $2x^3$-$5x^2$-$4x + 3 = 0$ (**10 marks)**

10 Figure 3 is a prism with cross section of an equilateral triangle PQR. PTST, PQVT and QRSV are rectangles such that RS = 12cm, QP = 6cm and X is a midpoint. Of QR.

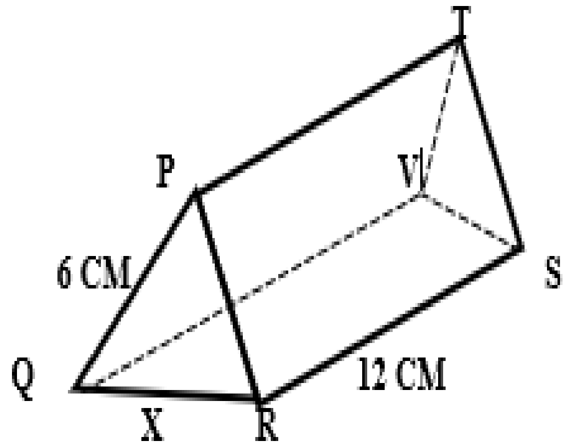

Figure 3

a. Calculate the length of TX giving the answer correct to 2 decimal places. (5 marks)

b. Calculate the angle between TX and the base QRSV giving the answer correct to 2 decimal places.

11 a. The distance (d) covered by a moving object is partly constant and partly varies with the square of its velocity (v). When the velocity is 4m/hr, the distance covered is 30m, and when the velocity is 6m/hr, the distance covered is 40m. If the distance covered is 72m, find the velocity of the object. (**6 mark)**

11 b. The initial speed of a car is 94km/hr. the speed increases by 24km every hour. If the car maintains the same pattern of speed, calculate the speed of the car in the 7^{th} hour. **(4 marks)**

12 A coffee brewer has 500kg Ricoffy coffee, 200kg of Mulanje coffee and 160kg of Black Ricoffy coffee from which she blends them to have two mixtures. Mixture A selling at k5 per kg contains 80% Ricoffy coffee, 30% Mulanje coffee and 20% Black Ricoffy coffee. Mixture B selling at k8 per kg contains 50% Mulanje coffee and 50% Black Ricoffy coffee.

a. If x represents mixture A and y represents mixture B, formulate three inequalities that satisfy the information in addition to $x \geq 0$ and $y \geq 0$. (3 marks)

b Using a scale of 2cm to represent 100 units on both axes, draw graphs to show the region represented by the inequalities on the graph paper provided. Shade the unwanted region. (5 marks)

c. Formulate the objective function hence find how many kilograms of each mixture should she blend to maximize her income. (2 marks)

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END OF QUESTION PAPER

NB: This paper contains 16 printed pages.